

Week 10 Lesson Plan

Enterprise RAG: The Library of Many Catalogues — Retrieval Architecture, from the Representation to the Verdict

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The Library of Many Catalogues

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Knowledge is of two kinds. We know a subject ourselves, or we know where we can find information upon it.

Samuel Johnson, 1775

Everything in the universe, and everything of man, would be registered at a distance as it was produced. In this way a moving image of the world will be established, a true mirror of his memory.

Paul Otlet, *Monde*, 1935

A Gentle Re-Entry

LAST WEEK THE CORPUS GOT ITS PAPERS. We read Google's young specification closely and learned that knowledge can carry its own provenance — who wrote it, who checked it, when it expires — and we seated that governed channel, honestly, at the very top of an escalation ladder we had not yet built. That was the gentle week; the breathing room was deliberate. Today we build the ladder rung by rung, and we take the whole day to do it: from eleven in the morning until half past eight tonight, this is the retrieval side of the house entire.

The seed was planted in June, on the very first day: retrieval is the load-bearing wall of RAG, because generation cannot cite what retrieval never surfaced. Every week since has leaned on that wall — the derivative artifacts of week four, the graphs of week five, the guardrails and evaluations that followed — and today the wall gets its engineering. The one thing to retain: *a retrieval system is not a search over documents. It is a portfolio of representations and a court of adjudication* — many purpose-built projections of one corpus, and a cascade of judges that decides, cheaply and then carefully, which projection to trust. Everything else today is that sentence, unpacked.

We will begin, as the monograph does, in Brussels in 1895, with a

This session draws on the RETRIEVALCRAFT monograph (*The Library of Many Catalogues*) and its deck. This lesson plan is your refresher and your first place to return; the monograph is the full anatomy; the book chapter will follow the session.

lawyer who catalogued the world by hand and built, a century early, exactly the architecture we are about to mechanize. Keep his image in mind all day. When the notation thickens — and by mid-afternoon it will — remember that every equation describes something Otlet's clerks once did with fountain pens.

What Must You Carry Forward

WHEN TODAY'S SESSION CONCLUDES, you should be able to:

- State the thesis — portfolio of representations plus court of adjudication — and explain, using the Mundaneum, why Otlet had the portfolio and lacked the court.
- Explain the recall ceiling with the bucket-and-sieve, and say precisely why no downstream component can raise it.
- State the tomographic principle and name what a single index claims that the portfolio refuses.
- Classify a query by its altitude (point, contextual, thematic, global) and name the representation each altitude needs.
- Contrast BM25, SPLADE, single-vector dense, Matryoshka, and ColBERT as lossy projections, each with the failure mode it prevents — and tell the BM42 story as a lesson in reading launches.
- Write InfoNCE and DCL, explain what leaves the denominator and why small batches stopped being a tax, and state the MRL-DCL invariant.
- Describe the hard-negative curriculum and the false-negative trap, and give the 48-hour fine-tuning doctrine with the case for refusing it.
- Recite the master pattern — retrieve the derivative, generate from the source — and name the two jobs of the provenance back-pointer.
- Walk the artifact inventory (rewrites, factoids, QA pairs, summaries, RAPTOR, community summaries, governed concepts) and for each name the corpse it prevents.
- Explain the long-context mirage — NoLiMa, context rot, the running meter — and the honest synthesis.

- Write reciprocal rank fusion with $k=60$, explain why ranks and not scores, and describe the surrogate-surface method for tuning what cannot be differentiated.
- State the which-text-to-rerank doctrine (the decoy's job ends at retrieval) and the parent-level dedup rule.
- Name the six query pathologies, describe the transformation cascade, place HYDE and the semantic cache, and say when not to retrieve at all.
- Give the version-skew doctrine, the ACL pre-filter rule, and the PoisonedRAG number.
- Apply the escalation ladder with the Shapley discipline, name the Kitchen Sink and the Sacred Cow, and explain why scale — not fashion — decides architecture.

If you can do these without hand-waving, you are ready for next week's semantic cache — the front door of the house we build today.

Act I: The Library — Foundations and the Portfolio

WE OPEN IN BRUSSELS, AND WE STAY THERE LONGER THAN YOU MIGHT EXPECT. The morning is foundations: the founding story and its exact load-bearing lesson, the thesis, the recall ceiling, the tomographic principle, and the four altitudes of a question. Then the first half of the portfolio — the representations themselves, sparse to late-interaction — and the instrument that produces them.

Brussels, 1895: The Mundaneum

TWO BELGIAN LAWYERS DECIDED TO CATALOGUE EVERYTHING. In September 1895 Paul Otlet and Henri La Fontaine convened the first International Conference on Bibliography and began the *Répertoire Bibliographique Universel*, a card catalogue meant to record every document humanity had produced. They rebuilt Dewey's decimal classification into the Universal Decimal Classification, whose connective signs let a single card sit at the intersection of *statistics : agriculture : India : 1895* — addressable from many directions at once. Then Otlet did the thing that concerns us: he decided the book was the wrong unit.

His *monographic principle* directed that documents be decomposed — each fact copied onto its own standard 3-by-5 card, classified, and filed where any future question could find it. By the 1930s the drawers held some sixteen million cards.¹

And people used it. From 1912 the Institute answered queries by telegraph. In 1934 Otlet's *Traité de documentation* sketched a *réseau* of desks and screens consulting a universal index over wires — a networked terminal, described the year Turing was an undergraduate. The world declined to cooperate: the government withdrew support in 1934; in 1940 the occupiers cleared the halls for an art exhibition, destroying some sixty-three tonnes of the collections. Otlet died in 1944 believing his life's work had failed. The surviving drawers eventually found refuge in Mons, where they stand today.

Here is the load-bearing lesson, and it is not “visionary anticipates the internet.” It is this: *Otlet did not build a search over documents. He built a portfolio of representations.* The same fact lived as a bibliographic entry by author, a classified card by subject-intersection, a clipping in a dossier, an image in the iconographic repertory, a line in an encyclopedic summary — five purpose-built projections of one source, each manufactured because some class of future question would need exactly that projection. Beneath them all, the monographic principle: decompose the document into atomic facts, because the question is almost never shaped like the book. And one thing was missing: *Otlet had no ranking.* A UDC class contained a card or it did not; within the drawer, cards sat in accession order. The Mundaneum had the portfolio and lacked the court. Half of today is the half Otlet never built.

The Thesis, and the Recall Ceiling

NOW CASH THE STORY INTO A SENTENCE YOU WILL CARRY ALL DAY. A retrieval system is two things joined: a **portfolio of representations** — many purpose-built, lossy projections of one corpus, each manufactured because some class of query needs exactly that projection — and a **court of adjudication** — a cascade in which cheap judges narrow the field so that expensive judges can afford to be careful. Retrieval architecture is the discipline of deciding which representations to manufacture and how to adjudicate among them. Scale decides how much of each you can afford; measurement decides which of them earn their place. One corpus, many catalogues.

Why is retrieval the load-bearing wall? Because of the **recall ceiling**. A child loses her marbles in a sandbox and recovers them with a bucket

¹ W. Boyd Rayward (1975). **The Universe of Information: The Work of Paul Otlet for Documentation and International Organisation.** Moscow: FID / VINITY

From 1912 the Institute ran a search service: queries by post or telegraph, clerks walking the drawers, answers back by mail at 27 francs per thousand cards — some 1,500 queries a year. Hear it with an engineer's ear: a remote client, a multi-index knowledge base, retrieved evidence with provenance, and a latency of days.

The two engines that let us run Otlet's architecture at machine speed: the LLM made the manufacture of derivative representations cheap (an iceberg of offline compute, invisible at query time), and sixty years of index structures made search sublinear. His institution died because every projection and every query cost a clerk; ours costs electricity.

and a sieve: scoop wide, scoop generously, tolerate sand — the sieve removes sand at leisure. But if the scoop misses a marble, no sieve can recover it. Whatever recall you have at the first stage is a ceiling that nothing downstream can raise: not a reranker, not a better prompt, not a bigger model. Generation cannot cite what retrieval never surfaced. This is why the scoop is tuned for recall and the sieve for precision, and why the entire cascade of Act III is a machinery for spending precision *after* recall has been secured. Write it on the inside of your eyelids: *only the scoop decides what is caught*.

The Tomographic Principle and the Four Altitudes

WHY MANY CATALOGUES RATHER THAN ONE BETTER ONE? A CT scanner never photographs the tumor. It takes hundreds of lossy projections — each blind on its own — and *reconstructs* what no single exposure contains. Each index in the portfolio is a projection of the corpus at its own semantic angle: the sparse lane sees the words on the page, the dense lane sees the meaning behind them, the factoid index sees the atomic claims, the summary index sees the theme. Fusion is the reconstruction. The kin are the cubist portrait — one man, many simultaneous angles — the compound eye, and *Rashomon*. What the principle is *not* is the panopticon: one eye claiming total sight. A single index makes that claim; the portfolio refuses it, and the refusal is the whole architecture in one gesture.

The projections are needed because questions arrive at different **altitudes**. A *point* query wants one fact ("what was the Phase III enrollment?"); a *contextual* query wants a passage with its surroundings intact; a *thematic* query wants what runs through a document ("how does this vendor think about data residency?"); a *global* query wants what lives in the structure of the whole corpus and in no passage at all ("what are the major regulatory concerns across these five hundred transcripts?"). Each altitude is served by a different representation — factoid, chunk, summary, community — and a query that lands on the wrong altitude gets a confidently irrelevant answer: the point query drowned in a summary, the global query answered from three quotes about one drug. Much of the routing and fusion machinery of the afternoon exists to send each question to its own height, or, better, to let the query *find* its own level.

The Instrument Cabinet: Every Representation Is a Lossy Projection

OPEN THE CABINET AND NAME WHAT EACH INSTRUMENT SEES. Korzybski's map is not the territory, and every index is a map drawn for one journey. The **sparse lineage** keeps a vocabulary-sized coordinate system: a document is a bag of weighted terms, and the inverted index makes lookup sublinear. BM25 weights by term frequency saturating against document length, and its two constants (k_1 and b) have survived thirty years because they encode a durable truth about how evidence accumulates. What BM25 cannot do is see *synonymy*: the query says "cardiac," the page says "heart," and the postings never meet. SPLADE answers by *learning* the expansion — a masked-language-model head predicts, for each document, a sparse distribution over the whole vocabulary, so the page about hearts carries weight on "cardiac" it never printed. It is a sparse lane that has read the thesaurus, and it keeps the inverted index and every operational virtue that comes with it.²

The **dense lineage** throws away the vocabulary coordinates and keeps a few hundred learned ones. A bi-encoder maps query and document into one space, and relevance is a cosine. This is the lane that closes the paraphrase gap — and it inherits the geometry of high dimensions, which we met in June and must now respect operationally.

Concentration of Measure and the Aperture Doctrine

IN A THOUSAND DIMENSIONS, ALMOST EVERY PAIR OF RANDOM VECTORS IS NEARLY ORTHOGONAL, and the cosine similarities of a real corpus, far from filling the interval $[-1, 1]$, crowd into a narrow band — typically somewhere between 0.2 and 0.9 depending on the embedder, with most mass in a sliver. This is concentration of measure, and it has three practical consequences you must carry into every threshold you ever set. First, an absolute similarity threshold is not portable across embedders: 0.7 on one model is a different aperture from 0.7 on another, because the distributions differ. Second, the *differences* between the top few candidates are small in absolute terms, so a reranker's discrimination matters more than the retriever's, and calibration at the exit is not optional. Third — the **aperture doctrine** — a threshold is a cone aperture around the query direction: widen it and you admit more sand; narrow it and you lose marbles. Set it on the sealed set, per embedder, per lane, and revisit it whenever the embedder changes. The Berlin experiment of Act II (0.64 versus 0.81–0.90 across a threshold of

² Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant (2021). "SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking". In: *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '21)*. Virtual Event, Canada: ACM, pages 2288–2292. DOI: 10.1145/3404835.3463098. URL: <https://doi.org/10.1145/3404835.3463098>

The two lineages are complementary, not competitive: the sparse lane wins on the rare term, the identifier, the exact phrase, the name that appears once; the dense lane wins on paraphrase and intent. This is precisely why hybrid retrieval, fused in ranks, beats either alone — the tomographic principle in its simplest, two-projection form.

0.7–0.8) is concentration of measure biting a real production system.

The Representations: Matryoshka, Late Interaction, and the Cost Knob

The BM42 story: a launch in 2024 claimed a new sparse method beat BM25 decisively; a careful reader recomputed the arithmetic within days and the claim did not survive. The lesson is not about one vendor. It is a reflex: read every launch the way you would read a paper you intend to replicate, and let your own sealed yardstick grade the homework.

³ Laxman Dhulipala et al. (2024). “MUVERA: Multi-Vector Retrieval via Fixed Dimensional Encodings”. In: *arXiv preprint arXiv:2405.19504*

⁴ Anthropic (Sept. 2024). **Introducing Contextual Retrieval**. <https://www.anthropic.com/news/contextual-retrieval>

TWO MORE IDEAS COMPLETE THE CABINET, and both changed the cost structure of the dense lineage rather than its accuracy — which is why they belong in an architecture session and not merely a modeling one. Then the two ideas that changed the cost structure. **Matryoshka representation learning** trains a single embedding whose prefixes are themselves good embeddings: the first 128 dimensions of a 1,024-dimensional vector are a usable, cheaper projection of the same meaning. Dimensionality becomes a runtime knob — scoop with 128 dimensions over millions of candidates, rescore the survivors at full width. And **ColBERT**’s late interaction refuses to pool at all: every token keeps its own vector, and relevance is the sum over query tokens of the maximum similarity to any document token (MAXSIM). The rare term that a pooled vector would average away keeps its vote. For years the objection was cost — a vector per token — and by 2026 the objection has largely fallen: MUVERA reduces multi-vector search to single-vector MIPS with guarantees, and WARP and small ColBERTs collapsed the serving bill.³ The tournament, not the cost objection, now decides where ColBERT sits.

The contextualization lineage closes the cabinet: Anthropic’s contextual retrieval prepends a generated situating context to each chunk before embedding and BM25, cutting top-20 retrieval failure by 35–67% depending on the reranker;⁴ late chunking and then whole-document contextual encoders moved that context from the prompt into the encoder itself. Watch that trend: some of the derivatives we manufacture by hand this afternoon are becoming the encoder’s native output.

The Trained Instrument: DCL, Curriculum, and the Invariant

THE EMBEDDER IS NOT THE GLASS; IT IS THE MIRROR, and mirrors are ground for a purpose. Two forces shape the geometry: *alignment* pulls a query toward its positive; *uniformity* spreads everything else across the sphere so that no region is crowded. InfoNCE implements both as a line-up tournament: for a query q with positive d^+ and a batch of N

candidates,

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(s(q, d^+)/\tau)}{\sum_{j=1}^N \exp(s(q, d_j)/\tau)},$$

where the denominator includes the positive itself. Read it aloud: make the positive win the line-up. The subtlety is that with the positive in the denominator, the gradient on the positive is throttled exactly when the model is already doing well — fifty times weaker, in the monograph’s worked case, precisely when it matters — and the fix is **decoupled contrastive learning**: evict the positive from the denominator,

$$\mathcal{L}_{\text{DCL}} = -\frac{s(q, d^+)}{\tau} + \log \sum_{j \neq +} \exp(s(q, d_j)/\tau).$$

The positive leaves the line-up, and small batches stop being a tax — the house loss for exactly that reason.⁵

Negatives arrive on a **curriculum**: easy random negatives first, then BM25-mined, then dense-mined, each stage stinging a little more, with distillation of a cross-encoder’s margins (Margin-MSE) as the load-bearing signal in serious recipes.⁷ And the two ideas of the morning meet in one rule: the **mrl-dcl invariant** — apply the decoupled loss *inside* the Matryoshka summation, at every prefix width, so that the 128-dimensional scoop and the full-width rescore were trained by the same discipline; and regression-test that invariant (recall at scoop depth on the sealed set) before any embedder swap, because a silently rotting scoop is six marbles gone before the sieve ever sees them. The 48-hour doctrine: with synthetic pairs manufactured from your own corpus, a curriculum, and DCL, a domain fine-tune is a two-day job — and you refuse it when the corpus is generic, the domain is small, or the yardstick does not yet exist to prove the swap honest.

Act II: The Second Corpus — Derivatives, and the Mirage

THE MORNING BUILT THE PROJECTIONS OF THE CORPUS WE WERE GIVEN. The early afternoon manufactures a *second* corpus — the derivative artifacts, Otlet’s cards mechanized — and then confronts the seduction that would abolish retrieval altogether: the million-token window. Both halves turn on the same question: whose shape is the text in, the reader’s or the examiner’s?

⁵ Chun-Hsiao Yeh et al. (2022). “Decoupled Contrastive Learning”. In: *arXiv preprint arXiv:2110.06848*. ECCV 2022

⁶ Gabriel de Souza P. Moreira et al. (2024). “NV-Retriever: Improving Text Embedding Models with Effective Hard-Negative Mining”. In: *arXiv preprint arXiv:2407.15831*

The false-negative trap: mine negatives too hard and the “negatives” are often unlabeled positives, and the loss punishes the model for being right. The countermeasure is positive-aware mining — discard candidates that score too close to the positive — and a denoising judge (a cross-encoder) that vets the hard negatives before they are trusted.⁶

The User Is the Examiner

AT IIT, I CAME SECOND IN APPLIED ELECTRODYNAMICS, and the manner of it has instructed me for forty years. The classmate who came first went to the seniors and collected the question banks: roughly forty questions this examiner actually asked, recycled with small mutations. He compiled beautiful answers to those forty, memorized them, topped the exam, and never opened a physics book again. I had optimized on learning the subject; he had optimized on *answering the examiner*. In retrieval architecture that is not a tragedy. It is the specification. *The user is the examiner*: a retrieval system faces a stream of questions in a register it does not control, and the corpus was written with no examiner in mind — textbooks for pedagogy, papers for reviewers, contracts for opposing counsel. That is the **register mismatch**, and it is permanent.

The orthodox response is to transform the query until it resembles the prose; we will do that too. But the deeper move runs the other way: *do not wait for the query to match your documents; manufacture documents that match your queries*. Take the corpus offline, when compute is cheap and time is abundant, and recast every chunk into the shapes a question can seize. Whatever intelligence you can spend on the still thing, spend it there — the iceberg of compute belongs below the waterline, invisible at query time.

⁷ Sebastian Hofstätter et al. (2020). “Improving Efficient Neural Ranking Models with Cross-Architecture Knowledge Distillation”. In: *arXiv preprint arXiv:2010.02666*

The Master Pattern and the Back-Pointer

ONE RULE GOVERNS EVERY ARTIFACT WE MANUFACTURE: *retrieve the derivative, generate from the source*. Rewrites, factoids, QA pairs, summaries, tree nodes, community summaries — these are *searched*. The original passage is what the generator sees and cites. The derivative exists to be found; the source exists to be quoted. A legal rewrite that dropped “irrevocable” — a term of art with case law attached — is not evidence; it is bait. And the load-bearing corollary: *every derived artifact carries a pointer back to its origin*, not as etiquette but as machinery. That **provenance back-pointer** does two jobs. It is the citation — the clickable, auditable claim — and it is the *invalidation key*: when a source changes, you walk the provenance graph in reverse and learn exactly which factoids die, which summaries have gone transitively stale, which tree nodes must be marked dirty. A derivative corpus without provenance is not an index; it is a rumor mill with vector search.

The Artifact Inventory

ONE SOURCE, MANY REPRESENTATIONS, EACH AGAINST A NAMED CORPSE. The discipline of the inventory is that no artifact earns its place by being interesting; it earns it by pointing at a specific way retrieval dies without it.

Contextualized rewrites — cure endophora and armored register (legalese, clinical shorthand). The chunk restated in the question’s register with pronouns resolved: the source’s understudy, auditioning for the retriever.

Factoids — cure *embedding dilution*. A five-claim paragraph embeds near the centroid of five meanings, far from every vertex; the pointed query lands near one vertex. The Berlin experiment: the paragraph containing the answer scores 0.64 against the river query, the extracted factoid 0.81–0.90 — and at a production threshold of 0.7–0.8, *the paragraph holding the answer is silently dropped*. Dense X industrialized this: FactoidWiki, 257 million propositions from 114 million sentences.⁸ Discipline: self-containment, an NLI faithfulness filter, a cap per chunk.

QA pairs — cure the cross-genre mismatch: the only artifact that puts a *query-shaped* object in the index. Three to eight questions per chunk; embed the question, carry the answer and source ids in the payload. The decoy the retriever *should* find.

Summaries — cure thematic queries shattering across chunks. Section, document, corpus altitudes, each its own index; the summarizer is a reporter, not a judge — itemize disagreement, never reconcile it.

RAPTOR nodes — the zoom lens. Cluster, summarize, recurse: a *discovered* hierarchy, $1.3\text{--}1.5\times$ the leaf count. Retrieve over the collapsed tree so the query finds its own altitude.⁹ Warning: garbage clusters yield confident garbage summaries — and the *phantom cluster*, an accidental clump that passes every per-chunk provenance check, is a provenance failure that only a topical-coherence check catches.

Community summaries — the telescope. Entities, Leiden communities, summaries at several resolutions, map-reduce at query time — for the sensemaking question that lives in no passage.¹⁰ Embeddings tell you passages *sound* alike; triplets tell you they are *about* the same things.

Return to Brussels: this part, entire, is the monographic principle in period costume. The cards are the derivative corpus. The shelved book is the source. The clerk retrieves the derivative and cites the original. Everything else is mechanization.

⁸ Tong Chen et al. (2024). **Dense X Retrieval: What Retrieval Granularity Should We Use?** Proposes proposition-level retrieval granularity. DOI: 10.48550/arXiv.2312.06648. arXiv: 2312.06648 [cs.CL]. URL: <https://arxiv.org/abs/2312.06648>

⁹ Parth Sarthi et al. (2024). “RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval”. In: *Proceedings of the 12th International Conference on Learning Representations (ICLR)*. Hierarchical summarization tree for multi-level abstractive retrieval. URL: <https://arxiv.org/abs/2401.18059>

¹⁰ Darren Edge et al. (2024). “From Local to Global: A Graph RAG Approach to Query-Focused Summarization”. In: *arXiv preprint arXiv:2404.16130*. Microsoft GraphRAG: knowledge graph construction via LLM extraction, Leiden community detection, and local vs. global search. URL: <https://arxiv.org/abs/2404.16130>

¹¹ Sam McVeety and Amir Hormati (June 2026). **How the Open Knowledge Format can improve data sharing**. Google Cloud Blog, June 12, 2026. The announcement post; frames OKF as formalizing the “LLM-wiki” pattern. URL: <https://cloud.google.com/blog/products/data-analytics> (visited on 08/08/2026)

The optical triad organizes the inventory: the factoid is the *microscope*, RAPTOR the *zoom lens*, GraphRAG the *telescope*. A well-run laboratory owns all three and knows which one it is holding.

Governed concepts — new this summer, and last week’s guest returned: the seventh artifact, refined until it stops being a projection and becomes an authored, typed, sourced, verified, expiring knowledge object. Its corpse is the **authority failure** — retrieval returning text that no one currently stands behind, the superseded policy at cosine 0.91. It is the only artifact whose derivation passes through a human review gate: extraction lands as a pull request; merge is the promotion event; repair is a diff, not a re-ingestion.¹¹

The Seventh Artifact: The Governed Concept, Seated

LAST WEEK’S GUEST RETURNS, AND TAKES A SEAT IN THE INVENTORY. The governed concept is the derivative artifact refined so far that it stops being a projection of the source and becomes an authored knowledge object — and it is worth pausing on the three properties that distinguish it from every other resident of the bestiary. First, *repair is cheap, and cheapness changes behavior*: a wrong fact in an embedded corpus waits for the quarterly re-ingestion; a wrong fact in a bundle is a one-line diff, merged in minutes, with git blame recording who fixed what. We rebuild every other artifact; this is the only one we *repair*. Second, it serves a second retrieval mechanism beside embedding — *navigation*: an agent walks the bundle’s index files and follows its links as a librarian walks stacks, a graph traversal over curated objects, kin to GraphRAG’s communities but with the graph *authored and reviewed* rather than induced by clustering. Third — carved over the door — *the review gate is structurally native*: the phantom cluster taught us that synthesis needs a coherence gate and we bolted one on; here the gate is the format’s own workflow.

The gate has a dark twin. An extraction agent can mint a concept whose every field is well-formed, whose every cited source is real, and whose definition is a fluent interpolation that no version of the policy ever contained — the **phantom concept**, the phantom cluster’s descendant, and more dangerous than its ancestor because a phantom that survives review is *promoted*: stamped human-reviewed, elevated into the high-stakes query paths the tiers exist to protect. The defenses are last week’s, aimed one stage earlier: the reviewer checks claim-to-source fidelity, not YAML hygiene; an entailment pass — the same NLI machinery that filters factoids — runs in CI on every knowledge pull request; and numbers get no prose at all, because a quantitative claim belongs in an attested computation where interpolation is structurally impossible. And the ladder discipline: the governed layer is the *top rung*, expensive

in sustained human attention, earned only for the governed core — glossary, metric definitions, policies, runbooks, the few hundred to few thousand concepts where the authority failure actually kills and a domain owner exists to hold the pen. An unreviewed bundle is *chunks wearing a suit*, and the suit makes it worse.

Two economics govern the inventory. The **sidecar doctrine**: the graph is a sidecar, never the main road — global sensemaking is 5–15% of enterprise traffic (our routers send one to two percent through it), disproportionately the questions executives ask, so it defines the intelligence ceiling without being allowed to carry the highway. The thousand-dollar intern invoice — an extraction job wired to the frontier API instead of the local cluster — taught the cost cliff; LazyGraphRAG and HippoRAG 2 have since repealed much of the invoice,¹² so the doctrine is now about *traffic*, not price. And the **archaeology of top-K**: log the genealogy of every final result for a week and count. The finding, every time: most of the top results, most of the time, are not raw chunks. That is the single strongest argument for the multi-representation architecture, and it costs one week of logging to run on your own traffic. Then keep it running as telemetry, and let genealogy — never enthusiasm — decide which lenses stay.

The channel's honest debit column: curation cost is real and recurring; *stale_after* is an alarm clock, not a maintenance crew; coverage is structurally partial; and the transformation itself can hallucinate. The wager that agents change the economics of curation is open — no production postmortems yet — and we say so.

The Economics of the Derivative Corpus

ENTHUSIASM FOR LENSES IS NOT A BUDGET. Raw chunks alone already carry four indices in the reference architecture (SPLADE, the 128-dimension scoop, full-dimension dense, ColBERT), and each artifact family arrives with indices of its own; the total passes a dozen without anyone noticing the odometer. So the discipline, stated as a rule and enforced as a gate: *every index must earn its place through ablation*. Turn each index off; measure the staircase on the gold set; an index whose absence nobody can measure is complexity without value. The house rule of thumb for the build-or-don't decision:

Artifact ROI $\approx$ (query-mix weight on that granularity) $\times$ (measured quality lift) $-$ (build cost + storage cost),

amortized over query volume. Factoids shine on fact-dense corpora with pointed traffic; rewrites pay only where a register gap exists; RAPTOR is overkill for a tiny FAQ; the graph sidecar needs sensemaking traffic to bill against. The weights come from your *measured* query distribution — the tyranny of the use case, in ledger form.

And the amortization has a denominator. “Work hard once, enjoy for the rest of your life” assumes the corpus sits still long enough to collect

¹² Bernal Jiménez Gutiérrez et al. (2025). **From RAG to Memory: Non-Parametric Continual Learning for Large Language Models**. arXiv:2502.14802. HippoRAG 2: passages as first-class PPR nodes; factoid parity with vector RAG plus wins on associative and sensemaking tasks.

the interest — and a regulatory feed that turns over weekly, or a ticket queue that turns over hourly, does not. The churn calculus is derivation cost per document, times churn rate, against quality lift, times query volume; when churn wins, build fewer, cheaper, lazier artifacts (Lazy-GraphRAG is exactly this trade, industrialized). When you do build, invalidation stratifies naturally: chunks, factoids, rewrites, and QA pairs invalidate *locally* (delete by document id, re-derive); summaries, tree nodes, and community summaries are *transitively* stale (mark dirty on write, rebuild lazily, bottom-up, asynchronously). Tiered freshness is not a compromise; it is the correct reading of how information decays at different altitudes.

Interlude: The Long-Context Mirage

Say's Law holds in retrieval: supply creates its own demand. Build the factoid index because the measured distribution was 80% point queries, and watch the thematic share climb as users discover the system can now be trusted with harder questions. Design for the distribution you will *create*, and leave room in the router for sidecars not yet built.

WHY RETRIEVE AT ALL, WHEN THE WINDOW HOLDS A MILLION TOKENS? The seduction is real and must be answered with evidence rather than doctrine. Needle-in-a-haystack benchmarks look solved because the needle shares words with the question; remove the lexical overlap — NoLiMa — and eleven of twelve frontier models lose half their accuracy by 32K tokens.¹³ Context rot is not a corner case; it is the default when the answer must be inferred rather than matched. Then the meter: a million tokens per question is dollars per question, every question, forever — the iceberg of compute dragged above the waterline and paid for at inference prices. And the library argument: you do not read the library to answer a question; you consult the catalogue, and Otlet's clerks — who *could* have read toward any answer — built the catalogues precisely because reading does not scale.

The honest synthesis: at cockroach scale — ten thousand documents, one register, a handful of candidates — the long window is a legitimate organ; stuff whole documents and let the reranker choose. Above that scale, retrieval is not competing with the window; it is what makes the window *worth filling*. The window is for holding evidence, not for finding it.

Act III: The Court, the Question, and the Machine Room

THE PORTFOLIO IS BUILT; NOW THE HALF OTLET NEVER HAD. The late afternoon convenes the court: how many candidates each lane emits, how the lanes are fused, how duplicates are collapsed, which text faces

the expensive judge, and how a verdict is made to mean something. Then the question side of the glass, the machine room where p99 lives, and the playbook that decides how much of any of it you can afford.

The Judicial Ladder and Fusion in Ranks

THE CASCADE IS A COURT IN WHICH CHEAP JUDGES NARROW THE DOCKET so that expensive judges can afford to be careful. The *scoop*: each lane — SPLADE, the 128-dimension Matryoshka lane, ColBERT — emits its top few hundred; recall is sacred here, and the depths are chosen so that the union rarely misses a marble. Then *fusion*. The lanes' scores are incommensurable — a BM25 score and a cosine share no scale — so we fuse in *ranks*, and the honest baseline is **reciprocal rank fusion**:

$$\text{RRF}(d) = \sum_{r \in \text{lanes}} \frac{1}{k + \text{rank}_r(d)}, \quad k = 60.$$

Read it: a document ranked highly by *any* lane rises; k damps the head so that one lane's first place cannot dominate; and $k=60$, from a three-author SIGIR paper of 2009, remains the default in essentially every engine.¹⁴ Learned fusion — tuned convex weights — beats RRF when you have the hold-out to tune it on,¹⁵ which brings the difficulty: fusion weights and emission depths sit on a *staircase* you cannot differentiate (a rank changes discretely). The house answer is the **surrogate-surface method**: sample the parameter grid, fit a smooth surrogate to the measured NDCG at each point, and optimize the surrogate — Bayesian optimization in a small number of dimensions, on the sealed set, with paired-bootstrap intervals so that a difference of 0.02 is not mistaken for signal.

The reference cascade, with numbers you may copy as a starting point and must then tune on your own yardstick: three lanes emit their top few hundred each (the sparse lane a little deeper, because it is the cheapest and its misses are the most embarrassing); fusion in ranks collapses the union to a few hundred; parent-level dedup and MMR bring it to a few dozen; a full-dimension rescore and, where earned, a ColBERT pass bring it to twenty or thirty; the cross-encoder reads those and returns its calibrated top ten to the generator. Latency at p95 stays under a second or two because every stage narrows before the next spends. Figure is the map to keep on your wall.

Dedup, the Decoy, and the Supreme Court

¹³ Ali Modarressi et al. (2025). “No-LiMa: Long-Context Evaluation Beyond Literal Matching”. In: *arXiv preprint arXiv:2502.05167*

¹⁴ Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher (2009). “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods”. In: *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759. DOI: [10.1145/1571941.1572114](https://doi.org/10.1145/1571941.1572114)

¹⁵ Sebastian Bruch, Siyu Gai, and Amir Ingber (2023). “An Analysis of Fusion Functions for Hybrid Retrieval”. In: *arXiv preprint arXiv:2210.11934*. ACM TOIS

Universal suffrage before the yardstick exists: until you have a gold set to tune on, RRF at $k=60$ with equal lane weights is the constitution. Amend it only with evidence.

BEFORE ANY JUDGE SPEAKS, ONE VOICE PER WITNESS. The derivative corpus guarantees that a single source will arrive many times — as its chunk, its factoid, its QA decoy, its rewrite — and a reranker handed five costumes of one passage wastes five seats. So collapse candidates sharing a parent *before* the sieve, then apply MMR for diversity among what remains. Then the doctrine that trips more teams than any other: *the decoy's job ends at retrieval*. When the cross-encoder arrives, it must judge the *source text* against the query — never the synthetic question, never the rewrite, never the summary. A reranker grading the decoy is a court cross-examining the bait. The supreme court itself — the cross-encoder that reads little and thinks hard — has become cheap enough (rerank-2.5, Rerank 4, instruction-following, 32K windows) that the question is no longer whether to seat it but how deep a docket to hand it; and its verdict must be **calibrated** at the exit, so that a score of 0.8 means the same thing on Tuesday as it did at launch and an abstention threshold can be trusted.

The Question Side of the Glass

HALF THE ARCHITECTURE STANDS ON THE QUERY'S SIDE. A query is a description of a gap in someone's knowledge, typed in a hurry, and it arrives broken in a small number of recurring ways. Six pathologies, six verbs:

Misspelled — *correct*. Cheap, deterministic, first; the sparse lane is unforgiving of a transposed letter.

Underspecified — *expand*. "The policy" becomes "the 2026 reserving policy"; pseudo-relevance feedback and the session's own context supply what the user left out.

Overloaded — *decompose*. Two questions wearing one question mark are split, retrieved separately, and recombined — the multi-query fan-out.

Ambiguous — *disambiguate against the corpus*. "Java" means the island in one corpus and the language in another; the corpus, not a dictionary, decides.

Register-mismatched — *rewrite*. The plain question restated in the corpus's dialect — the read-time twin of the write-time rewrite artifact.

Multi-hop — *plan*. The question whose answer requires an intermediate answer first is decomposed into a chain and retrieved iteratively — the seed of the agentic turn.

The transformation cascade applies these in order of cost, and its most expensive resident is HYDE — fabricate a hypothetical answer and embed *that*, a document-shaped probe for a document-shaped index.¹⁶

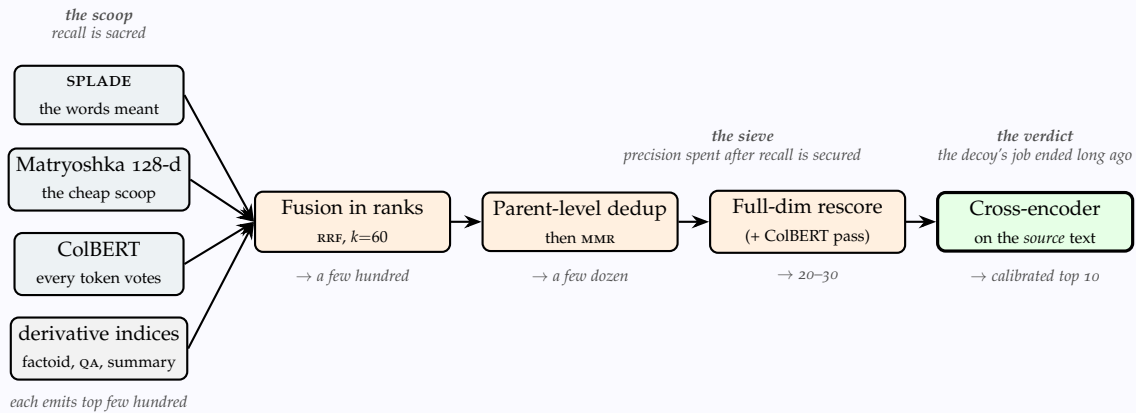


Figure : The reference cascade — a court in which cheap judges narrow the docket so the expensive judge can afford to be careful. Every stage narrows before the next spends; the cross-encoder reads only source text; the numbers are starting points to be tuned on your own yardstick.

It is generated QA pairs run at read time instead of write time: one principle, eliminate register mismatch by generating the missing side, at two timings. Use it last, with a capable model, knowing that a fabricated probe can fabricate a neighborhood.

Two doctrines from the derivative side return here in mirror image. The canonised query — spelled, expanded, disambiguated — is what every lane receives, so that the lanes vote on the same question; and the *original* query is what the cross-encoder judges against, so that a transformation which drifted from the user’s intent cannot launder itself through the court. Transform for retrieval; judge against the truth.

The Front Door, the Router, and the Agentic Turn

THE SEMANTIC CACHE is the front door: an embedder E and a threshold τ that decide whether this question has, in effect, been asked before — serve, hint, or miss, never guess — and it deserves its own day, which it gets next week. The **router** sends each question to the right catalogue and the right court: point and contextual traffic down the highway, thematic to the summary lanes, the global minority to the graph sidecar, schema questions to the structured store; and the agentic turn — Search-R1 and its kin, retrieval learned end-to-end by reinforcement¹⁷ — is the router’s most expensive tier, a rung on the ladder and not a heresy. And the quietest optimization: sometimes the right answer is not to retrieve at all — the parametric answer suffices, or the question is a

¹⁶ Luyu Gao et al. (2023). “Precise Zero-Shot Dense Retrieval without Relevance Labels”. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, pages 1762–1777. DOI: [10.18653/v1/2023.acl-long.99](https://doi.org/10.18653/v1/2023.acl-long.99). URL: <https://aclanthology.org/2023.acl-long.99/>

The transformation cascade is a pipeline of understanding run in order of cost: cheap deterministic fixes first (spelling, normalization, acronym expansion), then classification (which pathology, which altitude, which route), then the LLM-mediated rewrites, and HYDE last. Each stage should be able to short-circuit the rest — most queries need only the first.

greeting, or the cache has it — and a system that retrieves reflexively pays for evidence it does not use.

The Machine Room

RECALL IS BOUGHT WITH MILLISECONDS, and the exchange rate is set by the physics of finding a neighbor: graph indexes (HNSW) trade memory for speed, disk-resident indexes (DiskANN) trade latency for scale, and quantization — product, scalar, down to one bit — trades precision for a budget line, with the scoop-and-rescore pattern recovering what quantization loses. Three doctrines keep the room honest. **Version skew:** a query must never span embedder versions; migrate by dual-write, serve from one index at a time, and never compare a v2 query against v1 vectors. **Poisoning:** PoisonedRAG showed that five crafted documents in a corpus of millions can steer ninety percent of targeted answers,¹⁸ so treat every ingested document as untrusted code, spotlight retrieved evidence, and watch for silent retrieval collapse on polluted pools. **Access control:** sync ACLs at index time and *pre-filter* at query time, never post-filter — a post-filtered top-K that removes eight of ten results has silently become a top-two, and the recall ceiling has fallen without anyone noticing. The cost ledger — storage bytes and marginal dollars per index — is what the CFO reads, and every index in the portfolio must post to it.

Serving the middle judges is where most latency budgets are won or lost: the full-dimension rescore over a few hundred candidates is a matrix multiply that belongs on the same box as the index; the ColBERT pass, when earned, is where MUVERA's fixed-dimensional encodings and residual compression pay their way; and the cross-encoder's docket depth is the single knob that trades the most quality for the most milliseconds — tune it on the yardstick, watch it on the dashboard, and never let a demo's twenty-candidate docket silently become production's fifty. Index operations complete the room: the living catalogue must be re-derived on churn via provenance keys, migrated between engines by the dual-write playbook, and rebuilt from the corpus and the recipes when — not if — something is lost.

The Seeing Catalogue, Briefly

OTLET COLLECTED PICTURES TOO, and so must we. Two lanes: the two-tower lane (CLIP and its descendants) sees *style* — it embeds images and text into one space and cannot read a sentence in a diagram — and the

¹⁷ Bowen Jin et al. (2025). "Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning". In: *arXiv preprint arXiv:2503.09516*

¹⁸ Wei Zou et al. (2024). "PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models". In: *arXiv preprint arXiv:2402.07867*. USENIX Security 2025

caption lane manufactures textual derivatives of every image (Otlet's card, mechanized for pixels) so that the whole derivative machinery applies. Then the OCR-free turn: ColPali retrieves page *images* by late interaction, skipping the parser entirely,¹⁹ and the storage tax that once made this a luxury has been repealed by the same compression that saved ColBERT. The exhibit joins the same docket: the cascade does not care whether a candidate began as a paragraph or a page.

Coupling the Yardstick, and the Two-Column Ledger

NOTHING IN THIS SESSION IS DECIDABLE WITHOUT A YARDSTICK, and the evaluation weeks were the preparation for today. The gold set is sealed, versioned, and hidden from the engineers; every claim about a component is a paired-bootstrap interval on that set, never a point estimate; and public leaderboards are smoke tests, not instruments — Evalcraft's lesson, applied to us. The yardstick must be *living*: a gold set frozen at launch measures the users you no longer have, and Say's Law guarantees your users will change. Then Pacioli's discipline from Venice, 1494, applied to architecture: *every component posts twice*. Column one is the cost of presence — build steps, data demanded, latency, maintenance, the thing that can now break at two in the morning; vendors will itemize it for you eagerly. Column two is the price of absence — the query class that degrades without it, the metric that registers the degradation, the failure a user actually sees — and column two can only be written by measurement: an ablation, a tournament, an archaeology trace. A component with a full first column and an empty second column is not infrastructure. It is decoration.

The Playbook: Scale Decides, and Every Index Earns Its Place

AN OLD MAN UNDER HOUSE ARREST WROTE THE FIRST PAGE OF THIS PLAYBOOK. Galileo, in 1638: scale a bone up in every dimension and its strength grows as length squared while its weight grows as length cubed; the giant collapses under his own femur. Haldane's mouse walks away; the horse splashes. The insect wears its skeleton outside and thrives; the elephant cannot. *Scale decides architecture* — not taste, not fashion, not the latest post. At cockroach scale (ten thousand documents, one register) the *exoskeleton* — tuned BM25, a generous window, one cross-encoder — is not a compromise; it is correct. Scale up to heterogeneous registers and a regulatory quality bar and new failure modes *emerge*, in Anderson's exact sense: more is different, and

The clock budget, roughly: lanes in parallel, tens of milliseconds each; fusion and dedup, single-digit milliseconds; the full-dimension rescore, tens; the cross-encoder, one to several hundred depending on docket depth and model — unacceptable for a trading screen and free for a batch pipeline. Latency is not a performance metric; it is a structural constraint that decides which judges may sit.

¹⁹ Manuel Faysse et al. (2024). “Col-Pali: Efficient Document Retrieval with Vision Language Models”. In: *arXiv preprint arXiv:2407.01449*

The three reference architectures — small, mid, large — are fully specified operating points in the monograph: lanes, depths, rerankers, artifact sets, latency, and cost. The numbers are illustrative; the architecture is the invariant, tuned on *your* yardstick.

the *endoskeleton* — the portfolio, the derivative corpus, the court — is what the new physics demands.

Then the process that tells you when to *stop building*: the **escalation ladder**. The naive baseline — BM25 plus an LLM — is sacred; it is the null hypothesis, and every component you add is a claim that the null is insufficient. Build the baseline after the yardstick, never before; lock acceptability criteria in writing; escalate one component at a time, measuring at every rung; run tournaments; ablate ruthlessly. The rungs, for reference: L1 baseline, L2 dense lane, L3 hybrid fusion plus one cross-encoder (where quality typically takes its first jump), L4 late interaction, L5 the full cascade; around them the rings — L6 derivative sidecars, L7 pre-retrieval intelligence, L8 grounding loops, L9 the graph sidecar, and L10, the governed concept layer, earned only for the governed core and only where an owner will actually review. The ladder is a practical approximation to the **Shapley value** — a component’s expected marginal contribution averaged over all assembly orders,

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)],$$

the only fair division of credit, and the instrument that separates engineering from cargo cult.²⁰ Build-up (the ladder) samples one ordering; tear-down (ablation) samples another; together they approximate the fair ledger and expose the two named pathologies: the **Kitchen Sink** (every arXiv technique, adopted because it might help) and the **Sacred Cow** (the component kept because it was expensive or championed). If removing it changes nothing, the cow is beef. *Every index must earn its place through ablation.*

And the benediction the whole day has been building toward. What was destroyed in 1940 was not knowledge; it was *representations* of knowledge — catalogues, summaries, the cards themselves — and what made the loss irreversible was that Otlet’s derivation pipeline was made of clerks and forty years. Ours can be re-run. Every index we built today is a derivative, regenerable from two things: the corpus, and the recipes this session has written down. Burn our Mundaneum and we re-derive it — an iceberg of compute, a few days of pipeline time, a regression suite to prove the rebuild honest. *The corpus is the only irreplaceable artifact.* Guard it, project it many ways, and let the court decide.

Essential Readings

THESE ARE REQUIRED, and they are chosen so that each is a load-bearing wall of today's architecture — read them in this order.

- **Cormack, Clarke & Büttcher, *Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods***²¹

Three authors, one page, $k = 60$ — the constitution of the court until your hold-out set says otherwise. Read it to see how little arithmetic beat learned fusion for a decade, and why.

- **Chen et al., *Dense X Retrieval: What Retrieval Granularity Should We Use?***²²

Propositions as the retrieval unit — the monographic principle mechanized, with FactoidWiki's 257 million cards. The granularity argument in numbers; read it beside the Berlin experiment.

- **Yeh et al., *Decoupled Contrastive Learning***²³

The positive leaves the denominator. Short, and it explains why the house loss is what it is and why small batches stopped being a tax.

- **Sarathi et al., *RAPTOR***²⁴ and **Edge et al., *From Local to Global***²⁵

The zoom lens and the telescope. Re-read the GraphRAG paper from week five with the sidecar doctrine and the LazyGraphRAG cost correction in hand.

- **Modarressi et al., *NoLiMa***²⁶

The long-context mirage, measured: remove lexical overlap and the frontier halves by 32K. This is the evidence you will need the next time someone tells you retrieval is obsolete.

Oliver Twist's List of Readings

OPTIONAL, BUT EACH ONE DEEPENS A THREAD for those with appetite after a very long day.

- **Kusupati et al., *Matryoshka Representation Learning*** (2022)²⁷

Dimensionality as a runtime cost knob — the scoop's enabling trick, and the reason the MRL-DCL invariant matters.

²⁰ Lloyd S. Shapley (1953). "A Value for n -Person Games". In: *Contributions to the Theory of Games II*

²¹ Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher (2009). "Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods". In: *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759. DOI: [10.1145/1571941.1572114](https://doi.org/10.1145/1571941.1572114)

²² Tong Chen et al. (2024). *Dense X Retrieval: What Retrieval Granularity Should We Use?* Proposes proposition-level retrieval granularity. DOI: [10.48550/arXiv.2312.06648](https://doi.org/10.48550/arXiv.2312.06648). arXiv: 2312.06648 [cs.CL]. URL: <https://arxiv.org/abs/2312.06648>

²³ Chun-Hsiao Yeh et al. (2022). "Decoupled Contrastive Learning". In: *arXiv preprint arXiv:2110.06848*. ECCV 2022

²⁴ Parth Sarathi et al. (2024). "RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval". In: *Proceedings of the 12th International Conference on Learning Representations (ICLR)*. Hierarchical summarization tree for multi-level abstractive retrieval. URL: <https://arxiv.org/abs/2401.18059>

²⁵ Darren Edge et al. (2024). "From Local to Global: A Graph RAG Approach to Query-Focused Summarization". In: *arXiv preprint arXiv:2404.16130*. Microsoft GraphRAG: knowledge graph construction via LLM extraction, Leiden community detection, and local vs. global search. URL: <https://arxiv.org/abs/2404.16130>

²⁶ Ali Modarressi et al. (2025). "NoLiMa: Long-Context Evaluation Beyond Literal Matching". In: *arXiv preprint arXiv:2502.05167*

²⁷ Ajay Kusupati et al. (2022). "Matryoshka Representation Learning". In: *Advances in Neural Information Processing Systems*. NeurIPS 2022. DOI: [10.48550/arXiv.2205.13147](https://doi.org/10.48550/arXiv.2205.13147). URL: <https://arxiv.org/abs/2205.13147>

²⁸ Omar Khattab and Matei Zaharia (2020). “ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT”. in: **Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR ’20)**. Virtual Event, China: ACM, pages 39–48. DOI: 10.1145/3397271.3401075. URL: <https://doi.org/10.1145/3397271.3401075>

²⁹ Laxman Dhulipala et al. (2024). “MUVERA: Multi-Vector Retrieval via Fixed Dimensional Encodings”. In: *arXiv preprint arXiv:2405.19504*

³⁰ Anthropic (Sept. 2024). **Introducing Contextual Retrieval**. <https://www.anthropic.com/news/contextual-retrieval>

³¹ Wei Zou et al. (2024). “PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models”. In: *arXiv preprint arXiv:2402.07867*. USENIX Security 2025

- **Khattab & Zaharia, ColBERT (2020)²⁸ and Dhulipala et al., MUVERA (2024)²⁹**

Late interaction, and the theorem that reduced multi-vector search to single-vector MIPS — the cost objection, born and fallen.

- **Anthropic, Introducing Contextual Retrieval (2024)³⁰**

The highest-yield, lowest-glamour upgrade in the catalogue: a dollar per million tokens for a factor of three in retrieval failure.

- **Zou et al., PoisonedRAG (2024)³¹**

Five documents in millions, ninety percent steering — the reason the machine room has locks. Read beside the guardrails week.

- **Rayward, *The Universe of Information: The Work of Paul Otlet (1975)*³²**

The standard scholarly life of Otlet, and the reason we know the story at all. For the humanist in you: the man who built our architecture by hand.

A Calibration Note

THIS SESSION WAS LONG, AND IT WAS MEANT TO BE. We covered the Mundaneum and its exact lesson; the thesis, the recall ceiling, and the tomographic principle; the four altitudes; the representations from BM25 to ColBERT; the trained instrument with DCL, the curriculum, and the MRL–DCL invariant; the examiner and the master pattern; the artifact inventory through the seventh, governed artifact; the long-context mirage; the court — fusion in ranks, dedup, the decoy’s job, the calibrated verdict; the question side; the machine room; the seeing catalogue; and the playbook, from Galileo’s giants to the Shapley ledger. That is more than a day’s ideas, and you are not expected to hold all of it tonight. Hold the thesis and the recall ceiling. Return here for the rest.

Before the closing thought, the map the room reconstructed at the end of the day — which story taught which doctrine — because it is the fastest way back into this material when you return to it under time pressure:

The marbles in the sandbox — the recall ceiling: only the scoop decides what is caught.

The Mundaneum — one corpus, many catalogues; the portfolio without the court.

The CT scanner — the tomographic principle: fusion is the reconstruction; refuse the panopticon.

The classmate's forty questions — the user is the examiner; manufacture the derivative corpus.

The Berlin paragraph — embedding dilution, and the factoid that rescues the silently dropped answer.

The legal rewrite that dropped "irrevocable" — retrieve the derivative, generate from the source.

The thousand-dollar intern invoice — the sidecar doctrine: telescopes are not for wristwatches.

The million-token bid — the long-context mirage: the window holds evidence; it does not find it.

The scoreboard nobody could normalize — fuse in ranks; RRF at $k=60$.

The decoy on the witness stand — the decoy's job ends at retrieval; rerank the source.

The suspiciously cheap terabyte — the machine room's ledger; version skew and the pre-filtered ACL.

Galileo's giants — scale decides architecture; exoskeleton to endoskeleton; every index earns its place.

The card that came back — the governed concept, seventh artifact, top rung.

The stories were not decoration on the mathematics; they were the mathematics before it had notation. The marbles in the sandbox *are* the recall ceiling. The classmate's forty questions *are* the derivative corpus. The thousand-dollar invoice *is* the sidecar doctrine. Galileo's giants *are* the square-cube law of architecture. And behind all of them, a Belgian lawyer with sixteen million cards who had the portfolio and lacked the court — and whose card came back this summer, as typed markdown with its provenance in the frontmatter.

Scoop wide and sieve hard; manufacture what the examiner will ask; let cheap judges spend their pennies so the supreme court can afford its verdict; and let every index earn its place, or let it go. One corpus, many catalogues — and the discipline to know, by measurement, which of them to trust.

Next week: the semantic cache — the front door of the house we built today, and the place where the discipline of trust computed at read time from honest signals returns once more.

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